Merging Queries in Power BI

1. Introduction to Power Query Transformations

Power BI is a powerful business intelligence tool, and at the heart of its data preparation capabilities lies **Power Query Editor**. Power Query provides a wide range of data transformation operations that enable users to clean, reshape, combine, and organize data from different sources. After exploring transformations like importing data, cleaning, promoting headers, appending tables, renaming columns, and checking data types, the next major transformation is **Merging Queries**.

Merging is essential when different tables store information that is related but separated into different datasets. While **Appending** vertically stacks tables with similar structure, **Merging** horizontally combines tables based on a common column called a **Key**.

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2. Understanding the Need for Merging Queries

2.1. Real-world Scenario

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In most data systems—whether Excel, SQL databases, or enterprise applications—data is stored in multiple tables. Each table often contains different fields, but the tables are related through certain fields.

For example:

- A **sales order list** may contain basic order details like Order ID, customer name, order date, address, and shipping details.
- A separate **order details** table may contain product-level information such as department, category, product ID, discounts, profit, and sales value for each item in the order.

Each table answers different business questions:

- *Order List Table*: When was the order placed? Who placed it? Where is it shipping?
- *Order Details Table*: What was sold? At what price? What was the profit?

To analyze both aspects together—like *"Which cities generate the most profit?"*—we must first combine the relevant tables. This is exactly what **Merge Queries** enables.

3. Importing Data and Preparing for Merging

3.1. Loading the Excel Workbook

The user selects **Get Data → Excel Workbook** and chooses a file named **purchase_database.xlsx**. This workbook contains two sheets:

• Order List (Sheet 1)

Includes:

- Order ID
- Client Name
- Order Date
- Gender
- City
- State
- Shipping Date
- Sales Partner
- Logistic Partner

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• Order Details (Sheet 2)

Includes:

- Order ID
- Product ID
- Department
- Category
- Warehouse
- Sales Value
- Discount

- Profit

3.2. Identifying the Common Column

Before merging, you must identify **a common field**, which serves as the **key**. The Power BI system requires at least one column common between the two tables that contains matching values.

The column common in both tables is:

✔ **Order ID**

This is the *primary key* that links both tables together.

3.3. Why This Matters

Without a common identifier:

- Power BI cannot match which row from Table A corresponds to which row from Table B.
- A merge operation is not possible.

Thus, the presence of a shared column is mandatory.

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4. Opening Power Query Editor

After selecting both tables in the Navigator window, click:

Transform Data

This opens **Power Query Editor**, where merging operations are performed.

You should:

- Verify column headers
- Verify data types (e.g., text, number, date)
- Add descriptions or properties if needed
- Inspect row counts

4.1. Adding Table Properties (Documentation)

Power Query allows adding metadata through:

Right-Click Table → Properties

Here, you can:

- Rename tables
- Add descriptions such as:
    - "Order list table has 2,414 rows and contains basic order information"
    - "Order details table has 4,709 rows and includes product-level information"

4.2. Checking Row Counts

Using **Transform → Count Rows** provides clarity on:

- How many orders exist
- Whether Order IDs repeat

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For example:

- **Order List** has **2,414 rows** → 2,414 unique orders
- **Order Details** has **4,709 rows** → orders with multiple products

This difference indicates:

- Each order may contain more than one product, which is typical in transactional datasets.

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5. Understanding Merging Queries

5.1. Purpose of Merging

Merging is used when you want to:

- Combine columns from two tables
- Expand information horizontally
- Enrich one table's dataset with attributes from another table
- Perform relational modeling without using the Power BI Relationships View

5.2. When to Use Merge vs. Append

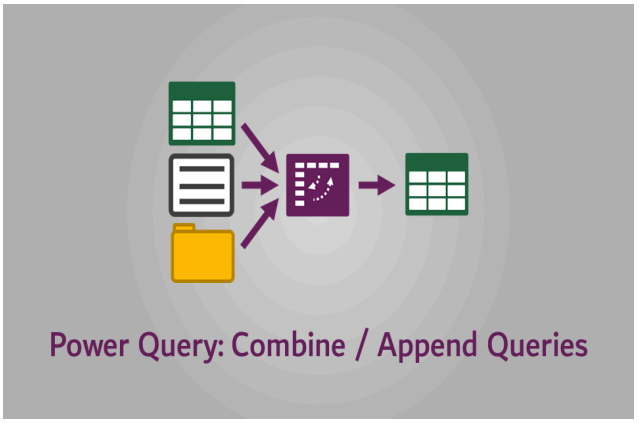
Operation	Purpose	Requirements
Append	Stack tables vertically	Same columns & structure
Merge	Join tables horizontally	Common key column(s)

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6. Performing Merge Queries

6.1. Merge Queries vs. Merge Queries as New

Power Query gives two merging options:



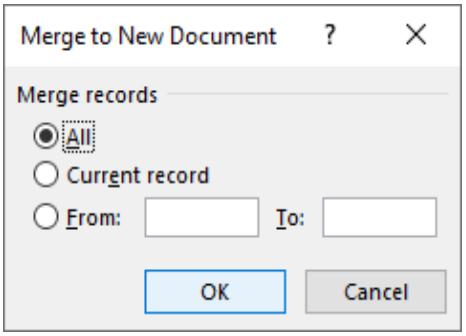
Option 1: Merge Queries

- Merges directly into the selected table
- Modifies existing query
- Best used when extending a master table

Option 2: Merge Queries as New

- Creates a separate, new table
- Keeps original tables intact
- Recommended for most analytical scenarios

7. The Merge Dialog Box – Understanding Components



The merge dialog box contains:

7.1. Upper Table (Left Table)

This is the first table selected—in our example:
✔ **Order Details**

7.2. Lower Table (Right Table)

The second table selected:
✔ **Order List**

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7.3. Selecting the Matching Columns

Click Order ID in both tables.

Power BI then analyzes how many rows match:

- Example Result: **4,708 of 4,709 rows matched**
    - Meaning: One row in Order Details does not have a corresponding Order ID in Order List.

This mismatch could be due to:

- Missing data
- Data entry errors
- Incorrect order creation flow

8. Join Types in Power BI Merge

Power BI supports **six join types**, similar to SQL joins and set operations.

Below is a clear explanation of each with interpretations:

8.1. Left Outer Join (Default)

✔ **Keeps all rows from the left table**
✔ Only matching rows from the right table

Result Includes:

- Every Order Detail record
- Only matching Order List records

Equivalent Set Theory Representation:
Left circle fully shaded + intersection area.

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8.2. Right Outer Join

✔ Keeps all rows from the right table
✔ Only matching rows from the left table

Good when:

- The right table contains the most important data to preserve.

8.3. Full Outer Join

✔ Keeps all rows from both tables
✔ Includes matching + non-matching rows

Used for:

- Audits
- Identifying mismatches
- Combining fragmented data

8.4. Inner Join

✔ Only keeps rows that have matching keys in both tables

Used when:

- You need strict intersections
- Missing data should be excluded

8.5. Left Anti Join

✔ Only rows from left table where no matching key exists in right table

Used for:

- Identifying missing data
- Finding orphan records
- Debugging data consistency

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8.6. Right Anti Join

✔ Rows in the right table not present in the left table

Useful for:

- Checking missing order details
- Auditing new or unmatched data

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9. Understanding Join Types Through Venn Diagrams

Power BI documentation uses Venn diagrams for intuitive explanation:

- Two circles represent tables
- Overlapping area represents matching rows
- Shaded area indicates rows included after merging



This approach is similar to set theory studied in mathematics and gives a clear visual cue.

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10. Completing the Merge Operation

Once the join type is selected (Left Outer in this case):

1. Power BI performs the merge
2. A new table named **Merge1** is created
3. The first column contains a expand icon
4. Clicking the expand icon allows selecting:
    - Columns to bring into the merged table
    - Whether to prefix column names

10.1. Row Count Verification

Even though only 4,708 rows matched:

- Left Outer Join preserves **all 4,709 rows** because:
    - Left table is Order Details

- It keeps all Order Details rows regardless of matches

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11. Working with the Merged Table

After merge:

- You may choose to expand related fields
- Remove unnecessary columns
- Rename merged columns
- Convert data types
- Perform further analysis

If merge was done incorrectly:

- You can delete the merged table
- Re-merge using a different join type

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12. Applying and Loading Merged Data

Once everything is ready, click:

Close & Apply

Power BI:

- Loads the data model
- Builds relationships automatically
- Makes merged data available for visualizations

This enables:

- Combined insights
- Calculated measures
- Advanced analytics
- Comprehensive reports

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13. Why Merging Queries Matters in Data Analytics

Merging allows you to:

- Integrate fragmented data
- Enrich datasets for better insights
- Create star schema models
- Build accurate reports
- Relate transactional and dimension data
- Maintain data integrity

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Common analytical use cases include:

- Joining customer demographics with transactions
- Combining product master with sales data
- Adding warehouse or logistic information to orders
- Merging region or territory info with performance metrics